AI IN GOVERNMENT AND PUBLIC SERVICES - VOLUME 8

The Data, Records and Interoperability Playbook

Ask once, find it fast, prove what happened - with AI that connects, finds and checks, and officers who own, retain and share the record.



Northbridge City and County holds the same resident in fourteen systems that do not agree, in paper files no search can reach, and in inboxes that were never records at all. This volume fixes the foundation every other volume assumed: one inventory, shared standards, lawful exchange, provable retention and access a citizen can actually use.

Gates D0-D5 | 8 Templates | 20 Golden Rules

HOW TO USE THIS PLAYBOOK

Start with the record, not the model

This volume is the plumbing volume, and plumbing decides whether the taps in Volumes 1 to 7 run clean or leak. Read it before scaling any AI that reads, summarises or shares public records. Every claim about Northbridge is fictional and illustrative; every rule is written to survive an auditor, a disclosure request and a coroner asking what was known, when.

How the volume is organised

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Core line: AI connects, finds and checks - officers own, retain and share the record.

EXECUTIVE LENS

Fund the foundation before the showcase. Interoperability is executive work, not IT housekeeping.

SERVICE LENS

Residents should never be couriers between your own departments.

ACCOUNTABILITY LENS

If you cannot produce the record, you cannot defend the decision made from it.

DIAGNOSIS

Fourteen versions of the same resident

Northbridge audited its holdings before buying anything. The findings will be familiar to any city or county.

Finding (illustrative)	Consequence
Same person in 14 systems, 6 spellings, 3 addresses	Officers re-ask, residents re-prove, errors compound
38 percent of case documents only in email or drives	No retention, no search, no disclosure readiness
Interfaces built point to point, undocumented	Every change breaks something nobody owns
Retention schedules exist; disposal rarely happens	Hoarded data, growing breach and disclosure cost
No catalogue: teams discover data by asking around	Duplicate collection, contradictory statistics

The diagnosis is not a technology gap. It is an ownership gap: records without owners, standards without enforcement, and sharing governed by habit instead of law. AI layered on this foundation does not fix it - it accelerates it, summarising the wrong version at scale.

> **Sequence matters:** inventory first, standards second, exchange third, AI fourth. Any vendor proposing the reverse order is selling the leak, not the tap.

WHERE AI PAYS

AI earns its place on retrieval, checking and connection

Use	Value	Guardrail
Semantic search across records	Officers find the file in seconds, not days	Permissions enforced before results, not after
Classification and metadata suggestion	Records filed completely at creation	Officer confirms; low confidence queues for review
Duplicate and mismatch detection	One resident, one reconciled view	Merge is a human decision with an audit trail
Extraction from legacy documents	Paper and scans become searchable records	Original retained; extraction marked as derived
Interface monitoring and mapping help	Broken exchanges caught early	No silent auto-remapping of meaning
Disclosure redaction support	Faster, more consistent responses	Officer reviews every redaction before release

Notice what is absent: AI deciding what a record means for a citizen, whether it may be shared, or when it may be destroyed. Those are accountability decisions, and this volume keeps them human on every page.

THE STOP LIST

What AI never does with public records

Stop list: no uncontrolled copies, no sharing without a lawful basis, no decision AI makes alone.

- **No uncontrolled copies.** No exporting records into personal drives, model training sets or vendor sandboxes outside the record system.
- **No sharing without lawful basis.** AI never approves a data share, a disclosure or a new use of a record.
- **No disposal by algorithm.** AI may flag records due for review; destruction requires the schedule, the owner and a signed authorisation.
- **No access decisions.** AI does not grant, widen or revoke who can see a record.
- **No silent merging.** Identity matches above a threshold are proposed, never applied automatically to merge citizen records.
- **No rewriting the record.** Summaries and extractions are derived layers, clearly marked, never replacements for the original.
- **No retention bypass.** Legal holds override every automated lifecycle rule, instantly and provably.

Post the stop list where interfaces are built, not only where policies are filed. An engineer at midnight is the audience.

PRINCIPLES

Six principles for data the public can trust

1. **One record, one owner.** Every dataset and record class has a named accountable owner.
2. **Original is sacred, derived is labelled.** Extractions, summaries and translations never masquerade as the record.
3. **Collect once, use lawfully many times.** Once-only works only inside purpose and consent limits.
4. **Standards before systems.** Shared identifiers, formats and vocabularies outlive any vendor.
5. **Permissions travel with the data.** Access rules are enforced at the interface, not remembered in a policy PDF.
6. **Prove it or it did not happen.** Every access, share, change and disposal leaves a retrievable audit entry.

EXECUTIVE LENS

Principles are budget decisions: fund owners, standards and audit before licences.

SERVICE LENS

A resident experiences principles as never being asked twice and never being surprised.

ACCOUNTABILITY LENS

An auditor tests principles by sampling records, not by reading the principles page.

INVENTORY

You cannot interoperate what you have not counted



Northbridge records archive: paper, scans and born-digital records brought under one inventory and one retention regime.

Build the inventory as a working register, not a one-off survey. For each system and record class capture: owner, purpose, lawful basis, personal and sensitive content, volume and growth, retention rule, quality state, interfaces in and out, and known duplicates.

How AI helps, honestly

Discovery tools can scan schemas, sample content, suggest classifications and flag likely personal data. Treat every suggestion as a draft for the owner to confirm. An inventory that is 90 percent auto-generated and never owned is fiction with better formatting.

> **Northbridge rule:** no new AI use case is approved against a dataset that is not in the inventory with a named owner. The gate is cheap; the alternative is not.

RECORD LIFECYCLE

From creation to disposal, every stage has a control

Stage	Control	AI role
Create / capture	File at creation, in the record system	Suggest class and metadata
Store	Access by role, encryption, backup proven	Detect anomalous access patterns
Use / share	Purpose-checked, logged, minimum necessary	Prepare extracts, flag over-broad requests
Review	Retention clock visible to owner	Queue due records for human review
Dispose / archive	Signed authorisation or transfer to archives	None beyond listing candidates

Email is the lifecycle failure mode. Northbridge rule: a decision communicated by email is filed to the record within two working days, with the thread, not a screenshot of it. AI filing assistants help; the duty stays with the sender.

STANDARDS

Shared meaning is the real interoperability

Systems exchange files easily; they exchange meaning badly. Northbridge therefore standardises meaning first.

- **Core vocabularies:** person, address, property, case, payment, organisation - defined once, versioned, published.
- **Formats:** open, documented formats for exchange and long-term preservation; proprietary formats carry an exit conversion plan.
- **Dates, codes and units:** one representation in exchange, however screens choose to display them.
- **Change control:** a standard changes by proposal, impact check and version - never by one team's quiet edit.

AI mapping tools can draft crosswalks between local and standard vocabularies. A human data steward approves each mapping, because a wrong mapping does not fail loudly - it lies quietly in every downstream statistic.

IDENTIFIERS

One person, one property, one case - carefully

Shared identifiers make once-only possible and make wrongful linkage possible too. Hold both truths.

Identifier	Use	Caution
Person reference	Link a resident across services they use	Probabilistic matches proposed, never auto-merged
Address / property reference	Permits, tax, land and services agree on place	Occupancy is not ownership; do not infer one from the other
Case reference	One case view across teams on the same case	Visibility follows role, not the identifier alone

Correction is a service: when a resident shows two records are wrongly linked, unlinking is fast, free, logged, and the correction propagates to every system that received the link. A linkage you cannot undo is a liability you have not priced.

INTEROPERABILITY

APIs with owners, contracts and consequences



Interoperability at Northbridge: governed interfaces with named owners, not a tangle of point-to-point extracts.

Every interface gets a published contract: what it provides, in which standard, at what freshness, under which lawful basis, with which owner on call when it breaks. Extract files emailed between teams are not interfaces; they are uncontrolled copies with a send button.

- Prefer query-in-place or event-based exchange over bulk copying.
- Version interfaces; never break a consumer silently.
- Log every exchange: who asked, what moved, under which basis.
- Monitor meaning, not only uptime: a live feed of wrong codes is a failure.

ONCE-ONLY

Ask once - inside the limits of the law

The once-only promise: a resident who has given Northbridge a fact should not be asked for it again by Northbridge. Delivered honestly, it has three limits printed beside it wherever it is promised.

1. **Purpose limit:** data collected for benefits is reused for tax only where law allows, not because the pipe exists.
2. **Freshness limit:** reused data carries its date and source; stale facts are re-confirmed, not silently trusted.
3. **Visibility limit:** the resident can see what was reused, from where, and correct it at the source.

AI pre-fill from held records is the visible win: forms arrive mostly complete, each field showing its source and date, each field editable. Acceptance of a pre-filled field is the resident confirming it - design the screen so that confirmation is real, not a pre-ticked assumption.

DATA QUALITY

Quality is measured at the point of harm

Count quality where a bad record hurts: a wrong address on an enforcement notice, a stale income figure in a calculation, a misspelled name on a licence. Northbridge tracks five dimensions per critical dataset: completeness, accuracy against source, timeliness, consistency across systems, and validity against the standard.

Fix at source, once

Corrections are made in the owning system and propagate outward. Downstream patching creates competing truths. AI anomaly detection queues suspects for stewards; it never overwrites a record to make a dashboard agree with itself.

Signal	Response
Duplicate rate rising	Review intake matching before merging anything
Cross-system mismatch	Steward adjudicates; source system wins by rule, not volume
Staleness beyond limit	Re-confirm with resident or downgrade reliance

METADATA AND CATALOGUE

Findable, understandable, requestable

The catalogue is the shop window of the inventory: searchable descriptions of what Northbridge holds, who owns it, what it means, how fresh it is, and how to request lawful access to it. If a dataset is not in the catalogue, it does not get shared, joined or fed to AI - because nobody can state its terms.

- Descriptions in plain language a new officer can understand.
- Sensitivity and lawful basis shown before any request is made.
- Quality scores and known limitations displayed, not buried.
- AI-generated descriptions and tags reviewed by the owner before publication.

> **Test:** give a new analyst one afternoon and the catalogue alone. What they cannot find, understand or lawfully request is the catalogue work still to do.

LAWFUL SHARING

Sharing is a decision with a name on it

Every share - internal, intergovernmental or external - rests on four recorded elements: the data, the purpose, the lawful basis, and the approver. Data sharing agreements name retention, security, onward-sharing bans and deletion duties for the receiver, and they expire.

Share type	Minimum control
Internal, same purpose	Role access, logging
Internal, new purpose	Compatibility assessment, owner approval
Other public body	Agreement, basis cited, minimum necessary
External / researcher	Agreement, de-identification review, secure setting
AI vendor processing	No training on records, sub-processor list, exit deletion proof

AI can draft the paperwork and check completeness. The approver reads, decides and signs - and that signature is itself a record, retained with the share.

RETENTION AND DISPOSAL

Keep what the law and history need - no more

Hoarding is not caution. Every record kept past its schedule enlarges breach impact, disclosure cost and the chance an outdated fact is used against a resident. Northbridge runs retention as a routine, evidenced operation.

1. Schedules set by record class, grounded in law and business need, reviewed on a cycle.
2. AI queues records reaching review dates with context for the owner.
3. Owner authorises disposal or archival transfer; legal holds freeze the queue instantly.
4. Disposal is executed across live systems, backups on their cycle, and vendor copies - with certificates.
5. The disposal log itself is retained: what, when, under which schedule, authorised by whom.

Rule: if you cannot prove disposal happened, plan as if the record still exists - because in a breach or a courtroom, it effectively does.

The record of the record

Some records outlive every system that created them: council decisions, land history, case precedents, the evidence base of public life. Transfer to archives is a designed stage, not a rescue operation from a dying server.

- **Provenance preserved:** creator, dates, versions, decisions taken from the record, and the chain of custody.
- **Authenticity checks:** fixity information verified on transfer and on a cycle thereafter.
- **Format strategy:** preservation formats plus rendered copies; migration logged as an archival event.
- **AI layers separated:** summaries, transcripts and tags transfer as derived material, labelled, never interleaved as original content.

Provenance is what lets a future reader - historian, auditor or resident - trust what they are reading. A record without provenance is a rumour in a nicer font.

LEGACY AND MIGRATION

Moving records without losing them or their meaning

Decommissioning is where records quietly die. Northbridge treats every system exit as a records project with a data budget, not an IT tidy-up.

1. Inventory the exit system first: what must survive, under which schedule, in what form.
2. Map to standards before moving; migration is the cheapest moment to fix meaning.
3. Reconcile counts and samples after migration, field by field for critical records.
4. Keep a read-only bridge until reconciliation is signed off, then close it on a date.
5. Record the migration itself: source, mapping version, losses or transformations, approver.

> **AI extraction warning:** OCR and extraction from legacy scans create derived data with error rates. Sample-check against originals, publish the error rate with the dataset, and keep the original image as the record until accuracy is proven for that class.

SEARCH AND RETRIEVAL

Find it in seconds - see only what you may



Access with privacy by design: the resident sees their record, the officer sees their remit, and both see an audit trail.

Enterprise search across records is the highest-value AI in this volume and the highest-risk if permissions are bolted on afterwards. The non-negotiables:

- Access control is applied at index time and query time; results a user may not open never appear, not even as titles or snippets.
- Ranking is explainable in plain terms: matched terms, date, record class, owning team.
- AI answers over records cite the exact record and passage; an answer without a retrievable citation is not shown to an officer as fact.
- Search itself is logged, because a search history is a record about the searcher and the searched.

DOCUMENT INTELLIGENCE

Extraction that knows it can be wrong

Classification, extraction and summarisation turn filing cabinets and inboxes into usable records - when their confidence is visible and their errors have somewhere to go.

Function	Design rule
Auto-classification	Confidence shown; below threshold, human files it
Field extraction	Each field links back to its source passage
Summaries	Labelled derived, dated, model version recorded
Translation	Original governs in any dispute; translation marked
Redaction suggestion	Officer confirms every redaction, item by item

Measure extraction accuracy per document class, publish it in the catalogue, and set reliance accordingly. A 97 percent field accuracy sounds good until the 3 percent is the date on a legal notice.

CITIZEN ACCESS

Your record, readable, correctable

Subject access is a service, not an obstacle course. Northbridge gives residents one route to see what is held about them, in plain language, with sources and dates, and one route to dispute it.

- Identity verification proportionate to sensitivity - strong, but never a barrier designed to deter.
- Explanations beside the data: what each item is, why it is held, who it is shared with.
- Corrections made at source and propagated, with confirmation listing where the correction went.
- Timelines met and evidenced; delays explained before the deadline, not after a complaint.

> **Design test:** a resident with no help, a phone, and twenty minutes can find, understand and challenge their record. If they cannot, the portal is a facade over the old process.

DISCLOSURE AND OPENNESS

Public records the public can actually get

Freedom of information, open data and proactive publication share one engine: records that are filed, findable and reviewable. AI helps most before the request arrives.

- **Proactive publication:** datasets and decision records published on schedule, in open formats, with methodology notes.
- **Request handling:** search support finds candidate records; officers decide exemptions and redactions, and reasons cite the record, not a template.
- **De-identification:** reviewed for re-identification risk in combination with other published data, not dataset by dataset alone.
- **Disclosure log:** what was released, when, in what form - itself published.

Open data is not a data dump. A dataset without definitions, freshness dates and known limitations invites confident misuse, and the correction never travels as fast as the chart.

AUDIT AND SECURITY

Every touch leaves a trace that survives scrutiny

Audit is the spine of accountability for records: who accessed, viewed, exported, shared, changed or disposed of what, when, from where, under which role. Logs are append-only, retained on their own schedule, and reviewed - an unread log is a hope, not a control.

- Bulk export and unusual-hours access alert a human reviewer the same day.
- AI anomaly detection prioritises review queues; it never clears its own alerts.
- Access recertification runs on a cycle; dormant access is removed, not renewed by default.
- Incident playbooks cover record breaches with resident notification duties and evidence preservation.

Rule: design every record flow as if its log will one day be read aloud in a hearing - because the important ones will be.

MEASUREMENT

Numbers that show whether the foundation holds

Measure	Healthy direction
Inventory coverage: holdings with named owner	Toward 100 percent
Once-only rate: facts re-asked of residents	Falling, by service
Time to retrieve a record for a case or request	Minutes, evidenced by sampling
Cross-system mismatch rate on critical fields	Falling, with adjudication backlog visible
Retention compliance: due disposals completed	On schedule, certificates filed
Access requests met in time, corrections propagated	High, with delay reasons published
Interface failures and silent data drift detected	Detected by monitoring, not by residents

Never measure volume shared as success. Sharing is a means; the outcomes are fewer re-asks, faster retrieval, defensible decisions and corrections that stick.

ECONOMICS

The foundation pays - count it honestly

Interoperability economics are mostly avoided cost and avoided harm, so count them in the open. Northbridge tracks: officer hours saved on retrieval and re-keying, duplicate system and storage costs retired, disclosure response cost per request, breach surface reduced by disposal completed, and the cost of standards, stewardship and audit that makes the savings real.

- Fund stewardship as a permanent line, not project residue.
- Charge interface building to the programme that benefits, catalogue and standards to the centre.
- Price the exit for every platform: export in standard formats, tested annually, costed in the business case.
- Do not book once-only savings until re-ask rates actually fall in sampled cases.

> **Honest claim:** the foundation rarely pays back in year one. It pays back every year after, in every service built on it - which is why it must be funded as infrastructure, not as a project with a closing date.

WORKFORCE

Stewards, archivists and owners are the product

Roles this volume depends on, named and staffed: data owners accountable per domain, data stewards who adjudicate meaning and quality, records managers who own schedules and disposal, archivists who own provenance, interface owners on call for exchanges, and a disclosure team that knows the holdings well enough to find anything.

- Train every officer in filing at creation - the cheapest quality control ever built.
- Give stewards authority to block a share or an AI use that violates a standard.
- Rotate officers through disclosure and corrections work; nothing teaches data quality like answering for it.
- Recognise invisible work: the best-run catalogue has no drama to report.

EXECUTIVE LENS

Protect steward and records roles in cuts; they are load-bearing, not overhead.

SERVICE LENS

Frontline staff file the record the resident will one day read. Train and tool them first.

ACCOUNTABILITY LENS

When something goes wrong, the named owner is the accountability. Vacant owner means vacant accountability.

FAILURE

How record programmes fail, in familiar ways

Failure	Early warning	Recovery
Big-bang migration	One cutover date for everything	Phase by domain, reconcile, keep bridges
Catalogue theatre	Hundreds of entries, no owners answering	Prune to owned, used datasets; rebuild trust
Search leaks	Snippets visible beyond permission	Halt index, fix at source, audit past queries
Hoarding by default	Disposal queue growing, holds abused	Executive review of holds, restart schedule
Standards drift	Local variants multiplying	Freeze new variants, fund convergence
Vendor lock of records	Export untested or charged as extra	Invoke contract, test export, plan exit

Every failure above was survivable when caught by measurement, and expensive when caught by a resident, a breach or a court. Instrument for the cheap catch.

SUPPLIERS

Buy platforms that respect the record

Procurement clauses that protect Northbridge records (see also Volume 6):

- Full export of records, metadata, audit logs and derived layers in documented open formats, tested before final payment and annually.
- No training on, or retention of, Northbridge records beyond the service; sub-processors listed and change-controlled.
- Retention and disposal executed by the vendor on instruction, with certificates, including at exit.
- Interface and standards compliance demonstrated against Northbridge vocabularies, not only vendor demos.
- Model or classification changes notified in advance with regression results on Northbridge samples.
- Audit log access for Northbridge reviewers without extra charge or delay.

> **Red flag:** a demo that searches everything beautifully but cannot say whose permission governed the results. That is not a search product; it is a breach with good design.

GATES D0-D5

Six gates from inventory to trusted exchange

D0 - Counted: inventory covers critical holdings; every dataset has a named owner or a closure date.

D1 - Described: catalogue live; standards and identifiers adopted for person, address, property and case.

D2 - Lawful: sharing register operating; bases recorded; vendor no-training terms signed.

D3 - Connected: priority interfaces on contract, logged and monitored for meaning, not only uptime.

D4 - Lifecycle: retention queue running; disposal evidenced; legal holds proven in a drill.

D5 - Trusted: search permission-tested, access and correction services meeting timelines, audit sampled independently.

Roadmap

Months 0-3: inventory and owners (D0). Months 3-6: catalogue, standards, sharing register (D1-D2). Months 6-12: priority interfaces and once-only pre-fill in two services (D3). Year 2: lifecycle at scale, permission-safe search, citizen record view (D4-D5). No gate is skipped because a later technology is exciting.

Fill-in templates: inventory, sharing, standards, retention

Template 1 - Dataset inventory entry

Dataset ___ | Owner ___ | Purpose ___ | Lawful basis ___ | Personal / sensitive content ___ | Volume and growth ___ | Retention class ___ | Interfaces ___ | Quality state ___ | Duplicates ___

Template 2 - Data sharing record

Data ___ | Receiver ___ | Purpose ___ | Lawful basis ___ | Minimum necessary check ___ | Security terms ___ | Expiry ___ | Approver signature ___

Template 3 - Standard / vocabulary change proposal

Term / code ___ | Current meaning ___ | Proposed meaning ___ | Reason ___ | Impacted systems ___ | Mapping ___ | Steward decision ___ | Version ___

Template 4 - Retention and disposal authorisation

Record class ___ | Schedule reference ___ | Date range ___ | Legal hold check ___ | Method ___ | Vendor certificates ___ | Authoriser ___ | Disposal log entry ___

Fill-in templates: access, interface, migration, correction

Template 5 - Citizen access and correction log

Resident ___ | Request date ___ | Identity check ___ | Records provided ___ | Correction claimed ___ | Source fix ___ | Propagation list ___ | Confirmation sent ___

Template 6 - Interface contract

Interface ___ | Provider / consumer ___ | Standard and version ___ | Lawful basis ___ | Freshness ___ | Owner on call ___ | Logging ___ | Breaking-change notice period ___

Template 7 - Migration reconciliation

Source system ___ | Record classes ___ | Counts before / after ___ | Field checks ___ | Extraction error rate ___ | Losses or transforms ___ | Sign-off ___

Template 8 - Assurance checklist

Inventory current? Permissions tested in search? Sharing register complete? Disposal evidenced? Holds drilled? Export tested? Audit sampled? Corrections propagated? Owner sign-off ___ Date ___

> **Assurance statement:** an independent reviewer can pick any record at random and trace its owner, basis, movements, retention state and audit trail within one working day. If not, the volume is not yet at D5.

20 golden rules for data, records and interoperability

1. Inventory before integration.
2. One record, one owner.
3. Original is sacred; derived is labelled.
4. Ask once, inside the law.
5. Standards before systems.
6. Permissions travel with the data.
7. No uncontrolled copies, ever.
8. Sharing is a signed decision.
9. Merge is human; linkage is reversible.
10. Fix at source, propagate the fix.
11. Disposal is evidenced, not assumed.
12. Legal holds beat every automation.
13. Search shows only what you may see.
14. An AI answer cites its record or stays silent.
15. Corrections propagate everywhere the error went.
16. Provenance is part of the record.
17. Test the export before you need the exit.
18. Measure re-asks, retrieval and mismatches - not volume shared.
19. Logs are read, or they are decoration.
20. If you cannot produce it, you cannot defend it.

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